11 PROPANE & HOT WATER

PROPANE TANK ARRIVING SOON

Propane hot water for the sinks and shower, with a solar pre-heater to save gas.

LEVEL	CREW	WHERE	STAGE NOW
■■■ Licensed hands connect	1–3	Near the outhouse/shower shed and the kitchen	Tank install underway; heater choice

A propane tank is being installed soon. It will feed a propane water heater for the sinks and the shower, and can also run the kitchen stove, a backup generator for the solar batteries (Guide 10), and a fire bowl for the lounge (Guide 05). Gas piping and connections are not a volunteer job: propane is heavier than air and pools at ground level, and this is a fire canyon. But volunteers can prepare the site, and build a solar pre-heater that does most of the work for free.

FIRST STEPS *today*

1. Yusuf: confirm with the installer where the tank and regulator go, and ask them to stub a line toward the water heater location (and any future generator or stove).
2. Decide the heater's location: close to the shower and sinks so the pipe runs are short.
3. Measure distances: tank to heater, heater to shower, heater to sinks. Note them on a sketch.
4. Volunteers: prepare a level gravel or concrete pad for the tank where the supplier wants it.
5. Volunteers: start the solar batch heater (below).

BEFORE YOU START

- **Volunteers do not cut, fit or connect gas pipe.** The propane supplier sets the tank and regulators; a licensed installer runs and pressure-tests the line, under permit.
- Smell gas (rotten eggs)? No flames, no switches. Shut the tank valve, walk away, tell Yusuf.
- Leak checks: soapy water brushed on joints. Bubbles mean a leak.
- A CO detector goes in any enclosed space with a gas appliance.

THE BASICS

Tank + regulators — A first-stage regulator at the tank and a second-stage one at the building bring the pressure down. The supplier sets the tank and its clearances.

Tankless — Heats on demand. A shower uses about 2 gallons per minute, a sink about 1. From cold well water (~60°F) to shower-hot (~105°F) is a 45°F rise; a 5–6 GPM unit covers a shower and a sink at once. It needs a little electricity (fan, igniter) and steady water pressure from the pump.

Tank heater — A small propane tank-type heater with an atmospheric vent can run with no electricity at all. Simpler off-grid; some standby heat loss.

One tank, many jobs — Hot water, stove, backup generator, fire bowl. Tell the installer every appliance you plan to add, so the tank, regulator and pipe are sized for the total load (BTUs) from the start.

Outdoor units — An outdoor-rated heater avoids indoor venting, but needs a noncombustible enclosure and a sheltered spot.

HOW-TO NOTES

Solar batch heater (volunteer project)	Strip an old water-heater tank to its steel, paint it flat black, and set it in an insulated, glass-fronted box facing south. Plumb it between the cold supply and the gas heater. Water arrives pre-warmed (often fully hot in LA summers), and the propane heater only tops it up. Free heat most of the year.
Even simpler	A coil of black pipe on a sunny roof feeding an outdoor shower.

THE PLAN *in order*

01	**Tank install**	Supplier sets the tank, regulator and clearances.
02	**Heater choice**	Tankless or tank; indoor or outdoor; size.
03	**Site prep**	Tank pad, heater enclosure (noncombustible), trench (depth set by the installer, tracer wire with the line).
04	**Install**	Licensed installer runs the line, pressure-tests, connects. Permit + inspection.
05	**Batch heater**	Built by volunteers, plumbed in as pre-heat.
06	**Check**	Leak test, CO detector, first hot shower.

TOOLS & MATERIALS

Volunteers: shovels, gravel, tamper, level, insulation, glazing, black paint, plywood/lumber for the batch box

Installer only: pipe, fittings, regulators, pressure gauge

FOR YUSUF TO DECIDE

- ? Tankless or tank-type heater
- ? Heater location
- ? Other propane appliances (stove, generator, fire bowl)
- ? Solar batch heater: yes or no

Resources: DigAlert: call 811 (digalert.org)